SIMULATED
Asset Management

# Concentration breach, remediation and verification

Agent: **Risk & Compliance**

**OPENING PROMPT**

```
Run a concentration check across all portfolios using the latest month-end holdings. Show me any positions above the 6.5% warning threshold, sorted by weight.
```

The demo date is the latest price date in the Snowflake Public Data share at build time, and positions are month-end. Agents answer from live data, so figures, names and rankings in your environment can differ from the examples in this script.

## Business Context Setup

**Persona**: James Chen, Compliance Officer at Simulated Asset Management
**Business Challenge**: Compliance officers run concentration checks across all portfolios every day, then have to track each breach through to resolution—retrieving breach history, finding the commitments portfolio managers made in engagement discussions, checking whether those commitments were executed, and reporting to the Risk Committee. The information spans holdings, policy documents, compliance alerts and engagement notes. **Value Proposition**: One conversation that moves from portfolio-wide scan to policy confirmation, breach history, PM commitments and a "promise vs reality" check, ending in a template-guided Risk Committee PDF.

**Agent**: `risk_compliance`
**Data Available**: Portfolio holdings across 11 strategies (positions are struck monthly, so each check runs against the latest month-end holdings), Concentration Risk Policy with specific thresholds, compliance breach alerts with resolution status, engagement notes with PM commitments, report templates

## Demo Flow

**Scene Setting**: James starts the day with a concentration check across all portfolios. The same breaches are on the agenda for the monthly Risk Committee, so he needs to confirm the policy, see which breaches are still open, find what the PMs promised and verify whether they delivered.

### Step 1: Portfolio-Wide Concentration Scan

**User Input**:

**PROMPT**

```
Run a concentration check across all portfolios using the latest month-end holdings. Show me any positions above the 6.5% warning threshold, sorted by weight.
```

**Tools Used**:

- `portfolio_analyzer` (Cortex Analyst) - Scan all portfolios for concentration

**Expected Response**:

- **Flagged Positions Table**: All positions >6.5% across portfolios
  - Portfolio name, Security, Ticker, Weight %, Market Value
- **Severity Classification**:
  - WARNING (6.5-7.0%)

    - BREACH (>7.0%)
- **Key observations**:
    - **Two names drive the breach list**: Costco and Goldman Sachs each breach in many portfolios — a firm-wide clustering problem rather than isolated positions
    - The largest single breaches are **Microsoft in SAM Global Balanced 60/40** and **Costco in SAM Global Flagship Multi-Asset**
    - The ESG-labelled portfolios (ESG Leaders, Renewable & Climate Solutions) carry Costco and Goldman Sachs breaches and are flagged for priority
    - A handful of WARNING positions sit in the 6.5-7.0% band

> ***Presenter Note**: The data genuinely breaches, so expect a long, populated list rather than a clean result. Lead with the Costco and Goldman Sachs cluster; AAPL and MSFT in SAM Technology & Infrastructure come back into focus at Step 4.*

**Talking Points**:

- **Portfolio-Wide Scan**: Single query checks all 11 portfolios simultaneously
- **Threshold-Based Flagging**: Automatic classification by severity level
- **Daily Efficiency**: Complete concentration check in seconds

**Key Features Highlighted**:

- **SAM_PORTFOLIO_VIEW**: Complete portfolio coverage for compliance monitoring
- **Automated Classification**: AI applies compliance thresholds automatically

## Step 2: Policy Threshold Confirmation

**Presenter Transition**:

> *"I see several flagged positions. Before I report these, let me confirm the exact policy thresholds to ensure I'm applying the correct limits..."*

*Reasoning: Policy thresholds may vary by portfolio type or have been updated. Confirming from source policy ensures accurate compliance assessment.*

**User Input**:

> **PROMPT**
>
> ```
> What are the exact concentration thresholds in our Concentration Risk Policy? Confirm
> warning and breach levels.
> ```

**Tools Used**:

- `search_internal_docs` (Cortex Search) - Retrieve concentration policy thresholds

**Expected Response**:

- **Policy Thresholds Retrieved** (Concentration Risk Management Policy):
    - Warning: **6.5%** single-issuer exposure — enhanced daily tracking and a documented hold/reduce rationale
    - Hard limit / breach: **7.0%** — mandatory reduction within 10 business days
    - Market-movement exception: a position pushed over 7.0% purely by appreciation must be cut below 6.8% within 20 business days, with no new purchases while above 6.5%
    - Policy section references for documentation

**Talking Points**:

- **Policy Verification**: Confirms thresholds from authoritative source
- **Audit Trail**: Documents policy basis for compliance decisions
- **Current Information**: Always retrieves latest policy version

**Key Features Highlighted**:

- **SAM_INTERNAL_DOCS**: Complete policy library with concentration requirements
- **Authoritative Source**: Policy retrieval ensures accuracy

**Follow-up — Breach Procedures**:

> **PROMPT**
>
> `What happens if we breach a concentration limit? What's the remediation timeline and what committees need to be notified?`

**Tools Used**:

- `search_internal_docs` (Cortex Search) - Breach procedures

**Expected Response**:

- **Remediation Timeline** (10 business days):
  - Day 0: automated alert to PM, CIO and Compliance
  - Day 1: written remediation plan from the PM
  - Days 1-10: rebalance to below 7.0%
  - Day 10: client notification if still unresolved
  - Day 15: regulatory filing if required
- **Governance**: the Investment Committee receives the concentration analysis monthly, the Board quarterly
- **Market-movement exception**: the 20-business-day path for appreciation-driven breaches
- **Severity note**: the agent links back to Step 1 — the live breaches mean the Day 0 and Day 1 obligations are already in play

**Talking Points** (follow-up):

- **Procedural Clarity**: Step-by-step breach response
- **Timeline Specifics**: Exact days and deadlines
- **Governance Integration**: Committee requirements explained

## Step 3: Retrieve Breach History and Resolution Status

**Presenter Transition**:

> *"The scan and the policy agree: these positions are over the limit. The Risk Committee will want to know how long they have been open and what has been done about them..."*

**User Input**:

> **PROMPT**
>
> `Show me all concentration breaches in the current monitoring cycle. Include which ones have been resolved and which are still active.`

**Tools Used**:

- `portfolio_analyzer` (Cortex Analyst) - Query the compliance alerts in SAM_PORTFOLIO_VIEW for breach history with resolution status

**Expected Response**:

- **Active Breaches**: led by Microsoft in SAM Global Balanced 60/40 and Costco in SAM Global Flagship Multi-Asset and ESG Leaders, with Costco and Goldman Sachs as repeat-offender issuers across funds
- **Active Warnings**: none outstanding — every active alert is a breach, which the agent reads as positions moving straight through the warning band
- **Resolved Breaches**: alerts closed this cycle with a resolution date, typically about a month after being raised, consistent with month-end rebalancing
- **Summary**: Counts of active versus resolved alerts

> ***Presenter Note**: Compliance alerts are raised against each portfolio's **mandate-specific limit** (5%, 6% or 8% depending on fund type), not the firm-wide 6.5% / 7.0% policy levels from Step 2. That is why some alerts trigger below 7.0% and why AAPL and MSFT in SAM Technology & Infrastructure are measured against 8.0%. The agent says this itself; if it does not, say it before the audience asks why the numbers differ from Step 1.*

**Talking Points**:

- **Complete Lifecycle View**: AI retrieves both active and resolved breaches from compliance database

- **Resolution Tracking**: Clear documentation of who resolved each breach and how
- **Duration Monitoring**: Days outstanding for accountability tracking

**Key Features Highlighted**:

- **SAM_PORTFOLIO_VIEW**: Comprehensive breach history with resolution status
- **Portfolio Analyzer**: Queries structured compliance alert data

## Step 4: Retrieve PM Commitments from Engagement Notes

**Presenter Transition**:

> *"Good, I can see active breaches requiring attention. Two of them sit in SAM Technology & Infrastructure, my most concentrated tech portfolio: Apple and Microsoft. Let me search our engagement notes to find what commitments the portfolio managers made on those..."*

*The presenter names Apple and Microsoft here: they are not the headline names in Steps 1 and 3, so the transition needs to introduce them.*

*Reasoning: Engagement notes document PM commitments made during compliance discussions. These commitments establish accountability and expected remediation actions.*

**User Input**:

> **PROMPT**
>
> ```
> Search our engagement notes for any compliance discussions or PM commitments regarding
> Apple and Microsoft concentration breaches.
> ```

**Tools Used**:

- `search_internal_docs` (Cortex Search) - Search for compliance discussion engagement notes

**Expected Response**:

- **Microsoft** (formal Compliance–PM remediation notes, all market-driven):
  - SAM Global Balanced 60/40: PM **Anna Chen** committed to reduce below the fund's limit via TWAP, with a target completion still in the future
  - SAM Technology & Infrastructure: Anna Chen committed to reduce below 8.0% towards a lower target weight, with a completion date that has **already passed** — the note's Investment Committee escalation clause is now engaged
- **Apple** (SAM Technology & Infrastructure):
  - PMs Anna Chen and Sarah Thompson committed to reduce below 8.0% towards a target weight, via TWAP with weekly updates
  - Several committed dates on file — the earlier ones have lapsed, the latest is days away
- **Risk flags**: overdue commitments triggering escalation, and a recurring pattern of both names drifting back above the limit

**Talking Points**:

- **Commitment Retrieval**: AI finds specific PM commitments from engagement notes
- **Deadline Awareness**: The agent compares committed dates with today's date and flags which have lapsed
- **Accountability Context**: Meeting dates, attendees, and specific promises documented

**Key Features Highlighted**:

- **SAM_INTERNAL_DOCS**: Searchable engagement history including compliance discussions
- **Compliance Discussion Templates**: Structured notes with PM commitments and timelines

## Step 5: Compare Current Positions Against Committed Targets

**Presenter Transition**:

> *"I have the PM commitments. Now the key accountability question—have they actually executed? Let me check current positions against those committed targets..."*

*Reasoning: Accountability requires comparing commitments to current reality. Portfolio data reveals whether promised reductions have been executed.*

**User Input**:

> **PROMPT**
>
> `What are the current weights for Apple and Microsoft in the SAM Technology & Infrastructure portfolio? Compare to the committed remediation targets.`

**Tools Used**:

- `portfolio_analyzer` (Cortex Analyst) - Query current portfolio weights from SAM_PORTFOLIO_VIEW

**Expected Response**:

- **Position Comparison**: both Apple and Microsoft remain **above the 8.0% limit and above their committed target weights** at the latest month-end, with no visible progress since the engagement notes were written

| Security | Committed Target | Current Weight | Status |
|---|---|---|---|
| Apple (AAPL) | 7.5% | 9.0% | Still in breach, over target, committed completion imminent |
| Microsoft (MSFT) | 6.7% | 8.2% | Marginally in breach, over target |

- **Remediation Gap Analysis**: no visible progress against the committed targets at month-end; the agent flags that missing the committed completion date would trigger the Investment Committee memo
- **Caveat**: positions are month-end, so any early-month TWAP execution would not yet show — the agent recommends confirming with the desk before escalating

**Talking Points**:

- **Promise vs Reality**: Clear comparison of commitments against actual positions
- **Accountability Gap**: Identifies unfulfilled commitments with specific shortfalls
- **Escalation Triggers**: Highlights positions requiring management attention

**Key Features Highlighted**:

- **SAM_PORTFOLIO_VIEW**: Latest month-end position weights for comparison
- **Accountability Tracking**: AI synthesises commitments with current data

## Step 6: Generate Risk Committee Report with PDF

**Presenter Transition**:

> *"I have the complete picture—breach status, PM commitments, and execution gaps. Let me retrieve the report template and generate a formal Risk Committee report..."*

*Reasoning: Risk Committee oversight requires formal reporting with standardised structure. The report template ensures all required sections are included.*

**User Input**:

> **PROMPT**
>
> `Find the Risk Committee compliance report template and generate a formal PDF report with all our findings on breach status, PM commitments, and remediation recommendations.`

**Tools Used**:

- `search_internal_docs` (Cortex Search) - Retrieve Risk Committee Compliance Report template
- `pdf_generator` (Generic) - Generate branded PDF with `internal` audience

**Expected Response**:

- **Template Retrieved**: Risk Committee Compliance Report template with required sections

- **Report Generated**:
  - Executive Summary: Active breaches and unfulfilled commitments
  - Compliance Status Overview: Active breaches table with severity, warnings and resolved items
  - Breach Details: Apple, Microsoft and the Costco / Goldman Sachs firm-wide cluster
  - Remediation Status: PM commitments (Anna Chen, Sarah Thompson), flagging Apple and Microsoft as overdue or behind schedule
  - Risk Assessment: Severity classification and trend analysis
  - Recommendations: immediate, near-term and monitoring actions with owners
  - Committee Actions Required: Approval checkboxes for escalation
- The month-end caveat from Step 5 is carried into the report
- **PDF Generation**: Professional branded PDF with SAM logo
- **Download Link**: Presigned URL for Risk Committee distribution

**Talking Points**:

- **Template-Guided Structure**: Report follows approved Risk Committee format
- **Comprehensive Coverage**: All required sections populated from analysis
- **Professional Deliverable**: Branded PDF ready for committee circulation
- **Audit Trail**: Complete documentation from breach to recommendation

**Key Features Highlighted**:

- **SAM_INTERNAL_DOCS**: Standardised report templates ensure consistency
- **Template + Data Synthesis**: AI combines template guidance with analytical findings
- **PDF Generation**: Professional branded documents for governance reporting

## Scenario Wrap-up

**Business Impact Summary**:

- **Time Savings**: Daily concentration check reduced from 30+ minutes to 5 minutes
- **Accountability**: Clear documentation of commitments vs execution
- **Committee Preparation**: Template-guided reports with professional PDF output
- **Gap Identification**: Automatic flagging of overdue remediation commitments
- **Audit Trail**: End-to-end documentation from breach to committee action

**Technical Differentiators**:

- **Portfolio-Wide Scanning**: Single query checks all portfolios simultaneously
- **Policy Integration**: Thresholds and breach procedures retrieved from authoritative policy documents
- **Three-Tool Integration**: Compliance alerts + engagement notes + portfolio data combined
- **Lifecycle Tracking**: From breach detection through PM commitment to execution verification
- **Template-Guided Reporting**: Standardised Risk Committee report structure
- **PDF Generation**: Professional branded documents for formal governance reporting